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FOURKITES

"From AI Awareness to Real Business Impact"

Abhishek Kumar

Abhishek Kumar is an AI expert and corporate trainer with ADaSCI. He is an engineering graduate from NIT Jamshedpur with over 10 years of experience. His expertise spans the entire data lifecycle, from data visualisation and analytics to advanced data science, ML, Agentic AI, and AI for business functions such as marketing, supply chain, HR operations, data security, SEO, and AEO.

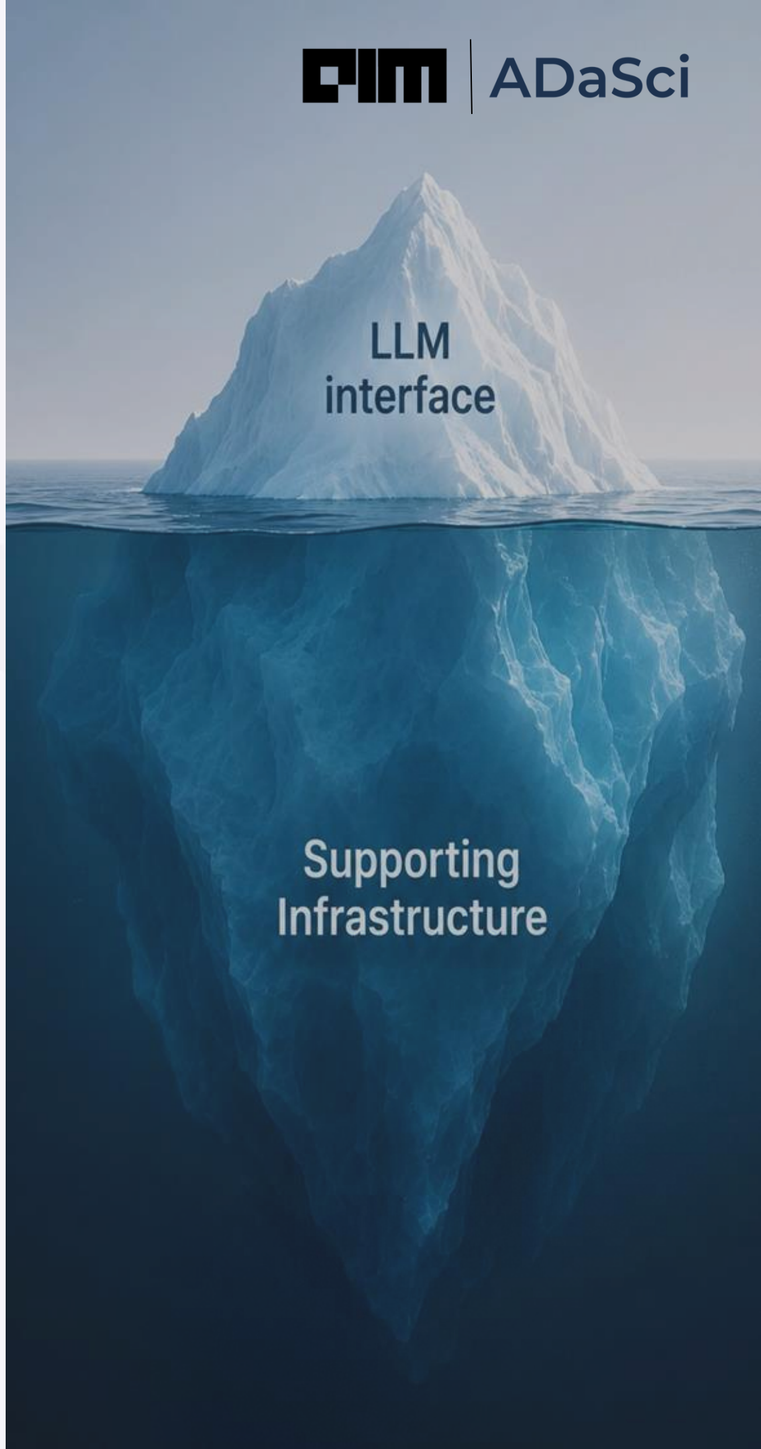
Abhishek excels at translating complex concepts into practical insights. He has a proven track record of empowering diverse audiences, delivering impactful trainings from Agentic AI Architecture for engineers to AI adoption strategies for leaders, including AI Adoption training for IOLCP, ICA training, and specialized tool-based programs such as SAS training for EXL.

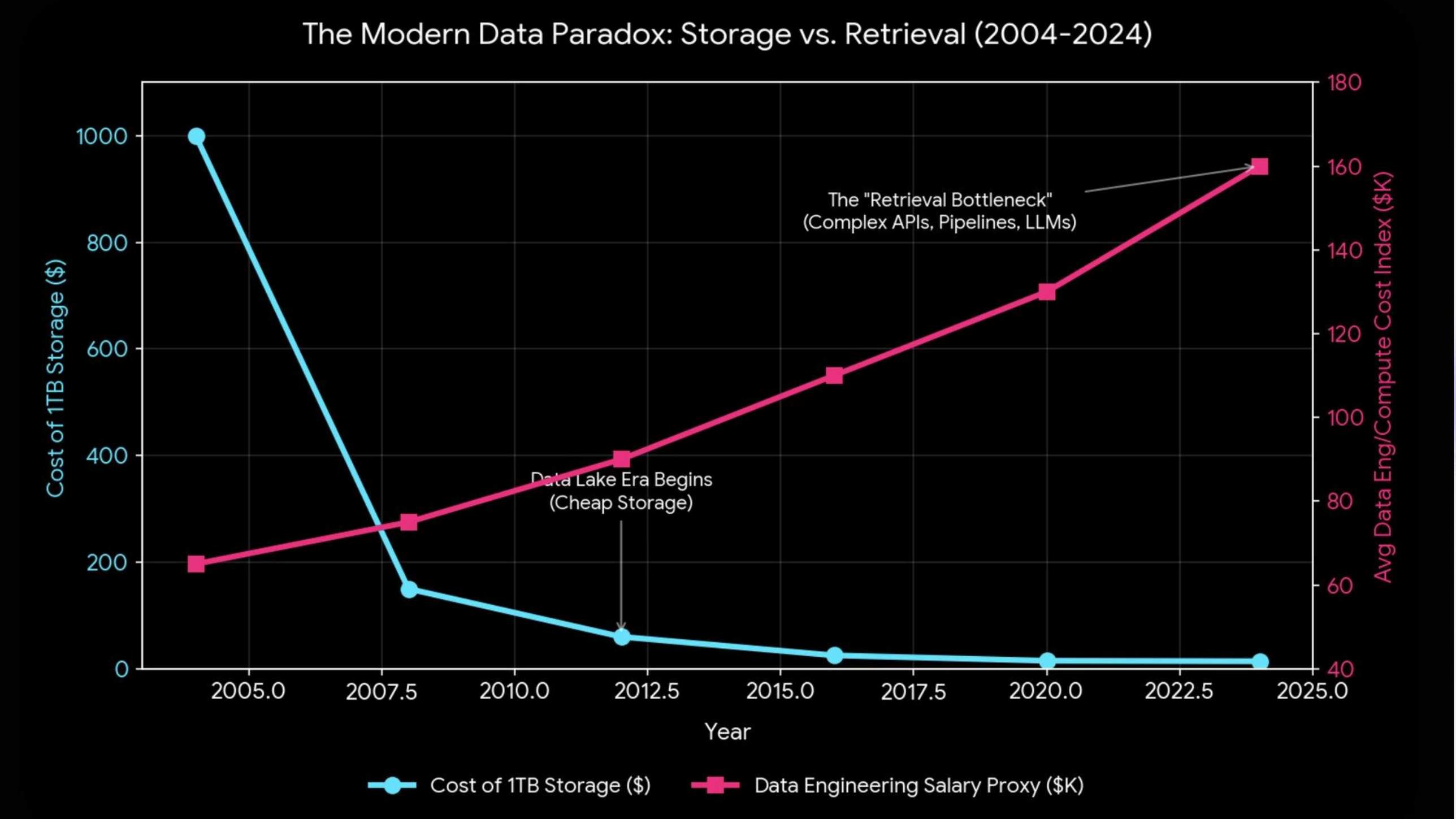
Data Pipelines for AI Systems

Behind Every Great LLM Application, Lies Exceptional Data Engineering

From fleeting interactions to continuous, contextual intelligence

May 2026




The Modern Data Paradox: Storage vs. Retrieval (2004-2024)
Cost of 1TB Storage ($)
Avg Data Eng/Compute Cost Index ($K)
Year
The "Retrieval Bottleneck"
(Complex APIs, Pipelines, LLMs)
Data Lake Era Begins
(Cheap Storage)
0
200
400
600
800
1000
40
60
80
100
120
140
160
180
2005.0
2007.5
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
2025.0
Cost of 1TB Storage ($)
Data Engineering Salary Proxy ($K)

1.1 Intelligence = Connecting Memory Points

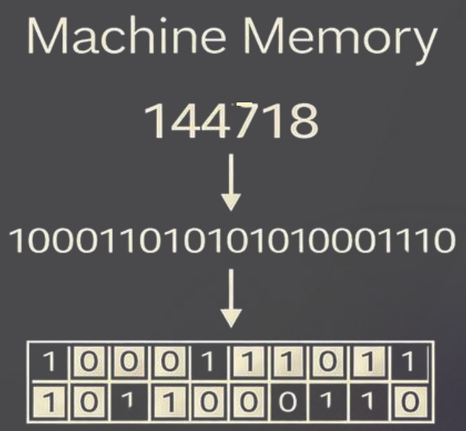
Human Memory

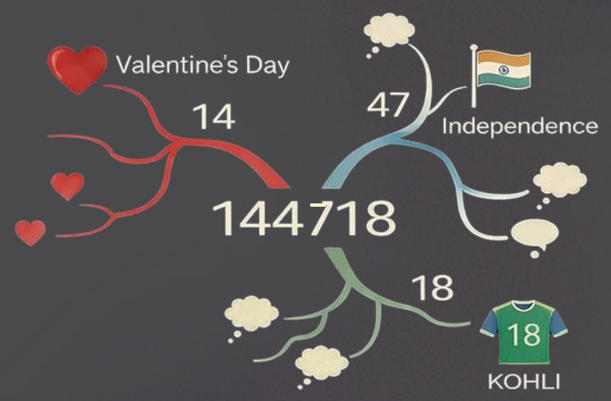
- Fluid and associative: Concepts linked through emotional weights and contextual relationships
- Inherently flawed but brilliant at synthesis: We reconstruct rather than retrieve
- No "file storage": We store concepts, not documents

Machine Memory (Traditional)

- Rigid and exact: Bit-perfect storage and retrieval
- Statelessness: Each prompt treated as isolated transaction
- The Memento problem: No continuity between interactions

The Paradigm Shift: True intelligence isn't just data storage, it is the effective connection of disparate memory points. Our goal is to bridge the gap by giving AI associative, context-aware recall.





2.1 The Three Memory Systems

Working Memory

The Context Window

"The immediate conversation." What the agent can access right now. Current dialogue, recent tool outputs, active reasoning context.

- Current conversation thread
- Active tool invocations
- Reasoning scratchpad

Episodic Memory

The Diary

"What happened, when, and with whom?" Chronologically ordered interaction logs capturing specific experiences and events.

- Conversation transcripts
- Tool execution history
- Timestamps & participants

Semantic Memory

The Encyclopedia

"What are the facts?" Core knowledge, rules, and learned user preferences abstracted from episodic experiences.

- User preferences & facts
- Domain knowledge
- Learned patterns & rules

Situational information like the next turn, pump in vicinity

Continuous record like dashcam feed, itinerary, bill records

Knowledge gained like hiking skill, information of mountains

3.1 Standard vs. Vector Database Comparison

Feature	Standard Database (SQL/NoSQL)	Vector Database
Search Type	Exact match (Keywords, IDs)	**Semantic match (Meaning, Context)**
Data Format	Rows, columns, JSON documents	**High-dimensional vectors (768-1536D)**
Query Logic	WHERE tag = 'apple'	**Find vectors closest to X**
Index Type	B-tree, Hash, Inverted	**HNSW, IVF, ANN algorithms**
Use Case	Transactional data, rigid records	**Similarity search, AI memory retrieval**

4.1 Similarity Metrics Explained

Cosine Similarity
Direction Only

Measures the angle between two vectors, ignoring magnitude. Perfect for text similarity where document length shouldn't matter.

- ✓ Best for: Text, documents of varying lengths
- ✓ Range: [minus one, one]
- ✓ Ignores: Vector magnitude (document length)

Dot Product
Direction + Magnitude

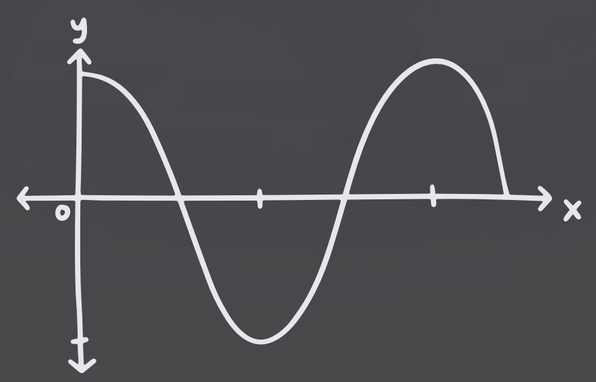
Measures both angle and magnitude. Useful when magnitude represents importance or quality (e.g., user activity levels).

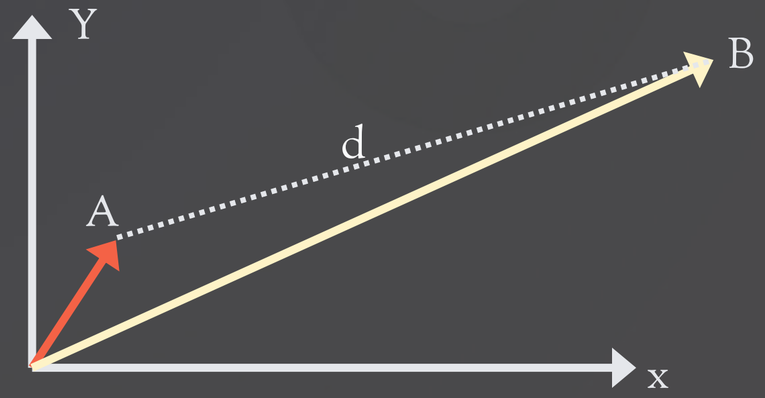
- ✓ Best for: Collaborative filtering
- ✓ Range: Unbounded (can be very large)
- ✓ Captures: Both similarity and importance

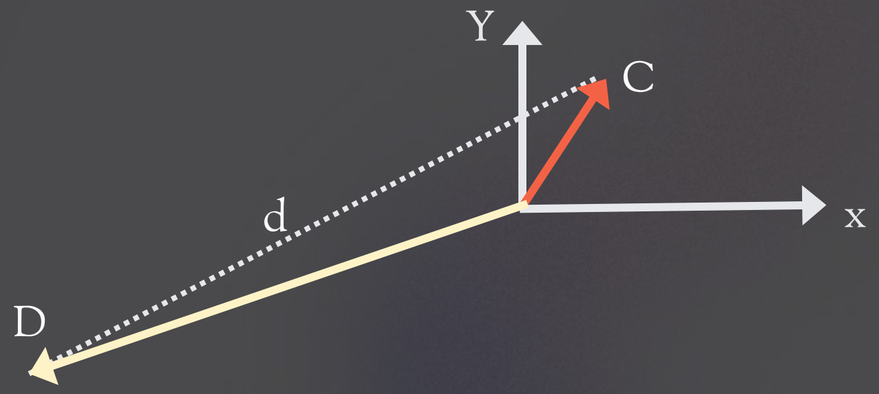
Euclidean Distance
Straight Line Distance

Measures straight-line physical distance between vector endpoints. Best when absolute differences in feature values matter.

- ✓ Best for: Clustering, anomaly detection
- ✓ Range: [0, ∞) where 0 = identical
- ✓ Sensitive to: Absolute magnitude differences







5.1 The Menace of Vectorizing Everything

Semi-Structured Database

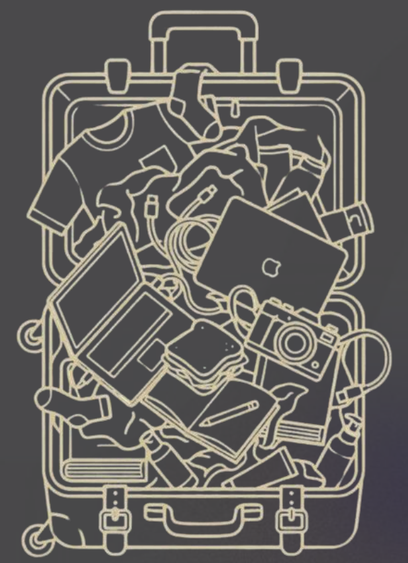
- Crisp, discrete records (e.g., a customer profile, a product description).
- Results into highly specific, accurate semantic vectors.
- The math works perfectly out-of-the-box.

Lengthy Textual Content

- Massive walls of text (e.g., 50-page PDFs, manuals, research reports)
- Requires processing before vectorization due to two roadblocks

Semantic Dilution: Combining multiple topics dilute meaning

Context Length Cap: Embedding models are often capped at 512 to 8,192 tokens



6.1 Chunking & Data Preparation Strategies

Fixed-Size Chunking

Best default for most use cases. Split text into fixed token sizes with overlap to prevent mid-sentence cuts.

Recommended Settings:

Chunk size: 400-512 tokens
Overlap: 50-100 tokens
(10-20%)

Structural Chunking

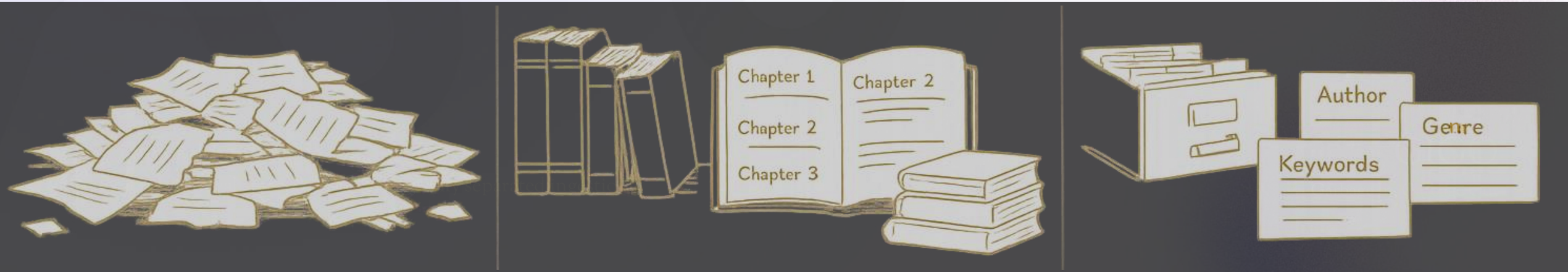
Respect document structure. Split by natural boundaries. Markdown headers, paragraphs, bullet points, code blocks.

- By headers (H1, H2, H3)
- By paragraphs
- By code blocks
- By bullet points

Metadata Tagging

Enrich chunks with context. Attach subheadings, dates, document IDs, section info before embedding.

- Section titles
- Document source ID
- Timestamps
- Content type tags



7.1 Time-Weighted & Hybrid Retrieval

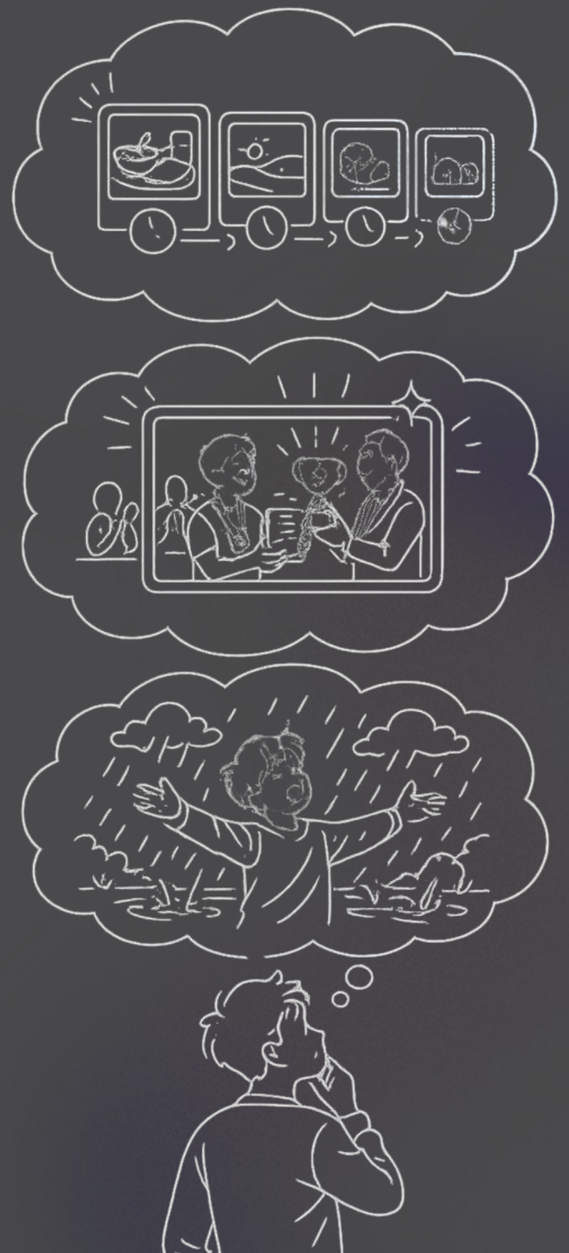
Stanford Generative Agents Research: Without proper retrieval scoring, agents treat all memories as equally relevant, leading to poor decision-making.

Stanford's Three-Dimensional Scoring

$$\text{Score} = (w_recency \times R) + (w_importance \times I) + (w_relevance \times Rel)$$

R = Recency	I = Importance	Rel = Relevance
Exponential decay over time	LLM-rated significance (1-10)	Cosine similarity to query

- Relevance finds matches, importance ensures critical facts always surface
- Weights may be fixed or adaptive depending on system design
- Importance score is pre-assigned during storage, not computed at retrieval



8.1 Context Pollution & Catastrophic Forgetting

Context Pollution

The "Lost in the Middle" Problem

The Issue: Shoving too many retrieved vectors into the prompt. The LLM gets confused, hallucinates, or succumbs to the "Lost in the Middle" phenomenon.

Research Findings (Stanford 2024)

U-shaped attention curve, beginning and end prioritized

The Fix

- ✓ Strict top-K limits: Retrieve only 3-5 most relevant chunks
- ✓ Strategic ordering: Place most critical info at start/end

Catastrophic Forgetting

Knowledge Overwrite

The Issue: When an agent learns new semantic facts, they can aggressively overwrite foundational knowledge. The agent "forgets" critical system behaviors.

Example Scenario

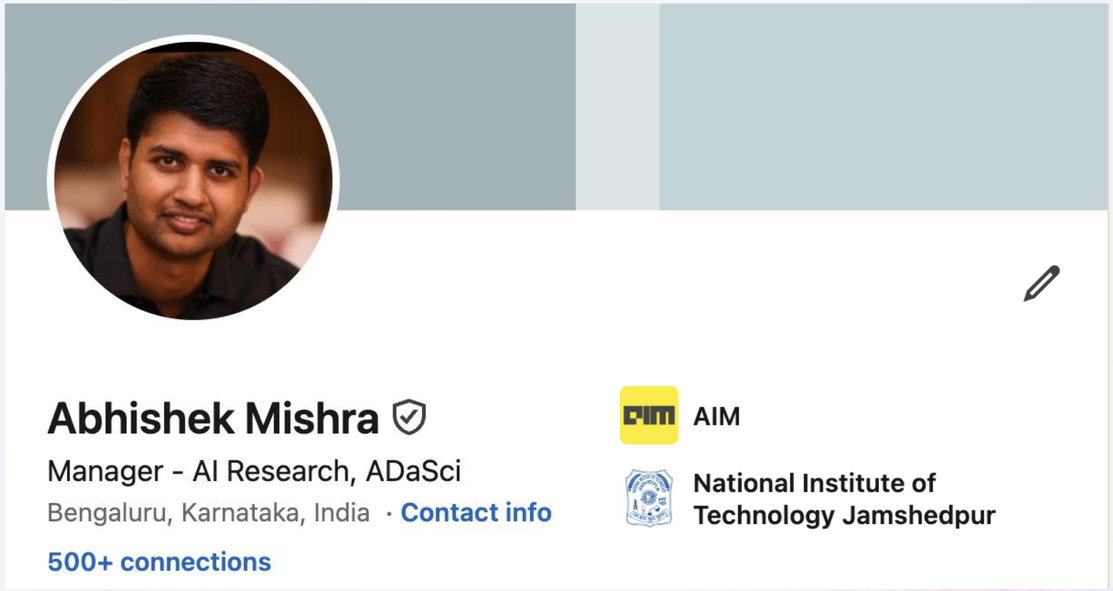
User frequently asks for quick access → Agent learns to skip verification

The Fix

- ✓ Segregated knowledge bases: Core vs. Learned separation
- ✓ Confidence thresholds: Require high certainty for overwriting

"When everything is important, nothing is"!

LinkedIn


Abhishek Mishra
Manager - AI Research, ADaSci
Bengaluru, Karnataka, India · Contact info
500+ connections
AIM
National Institute of Technology Jamshedpur

thank you